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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

GENERATIVE AI IN BIOINFORMATICS, MEDICINE AND HEALTHCARE

Chapter 6: Explainable AI

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Content

6.1. Introduction to Explainable AI

6.2. Explainable AI in GenAI

6.3. Several interpretation methods can be applied to Gen AI

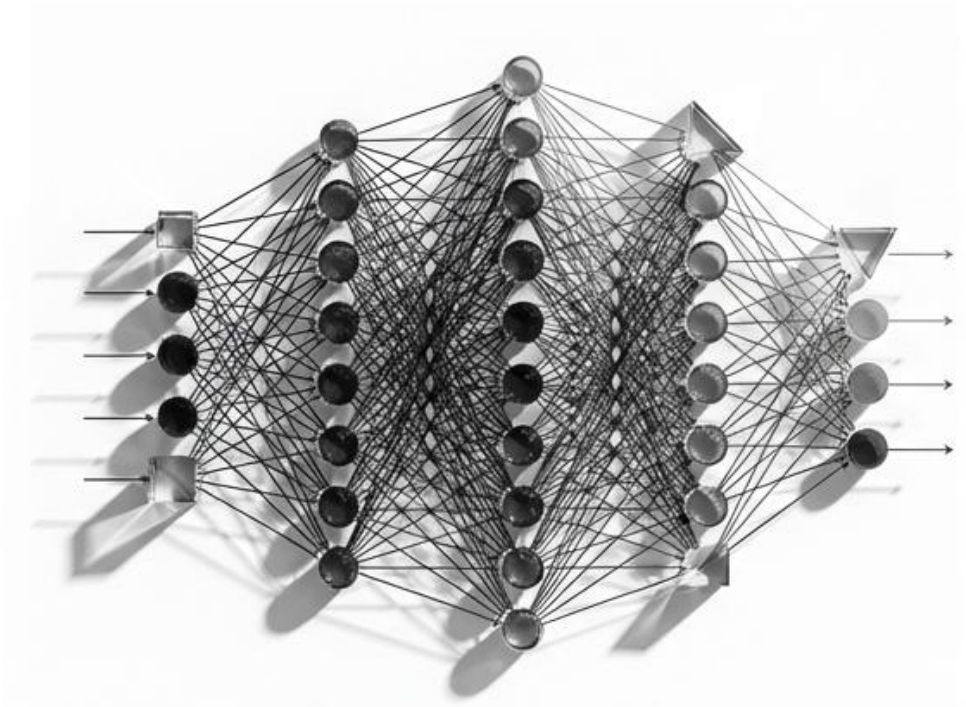
6.4. Evaluate interpretation by incorporating experimental data

6.5 Identify key biological factors in AI models

6.6. Interpretation in modern problems

Does it work right?

The Power



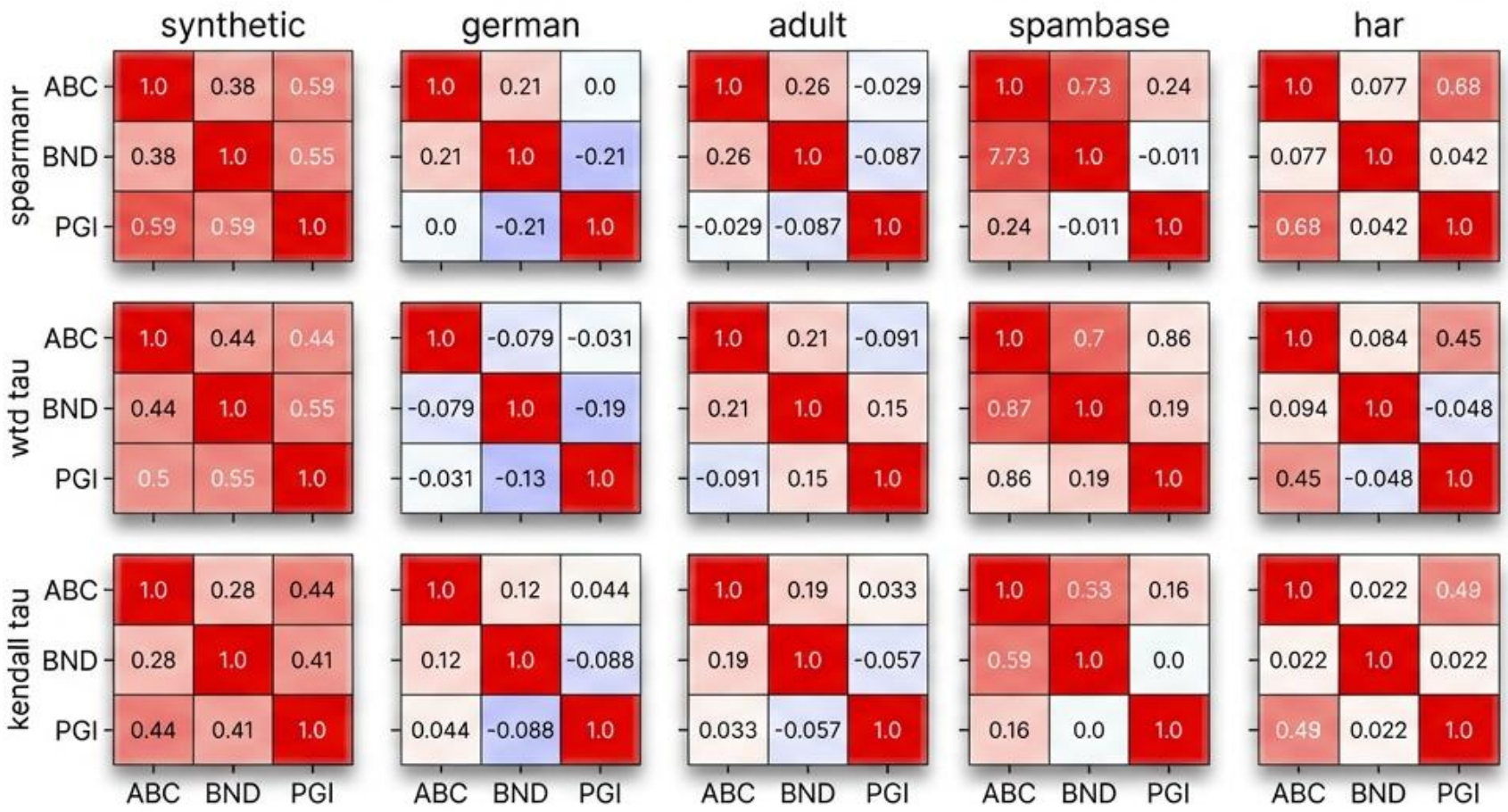
High Accuracy / Deep Learning

The Problem

Low Trust / Black Box

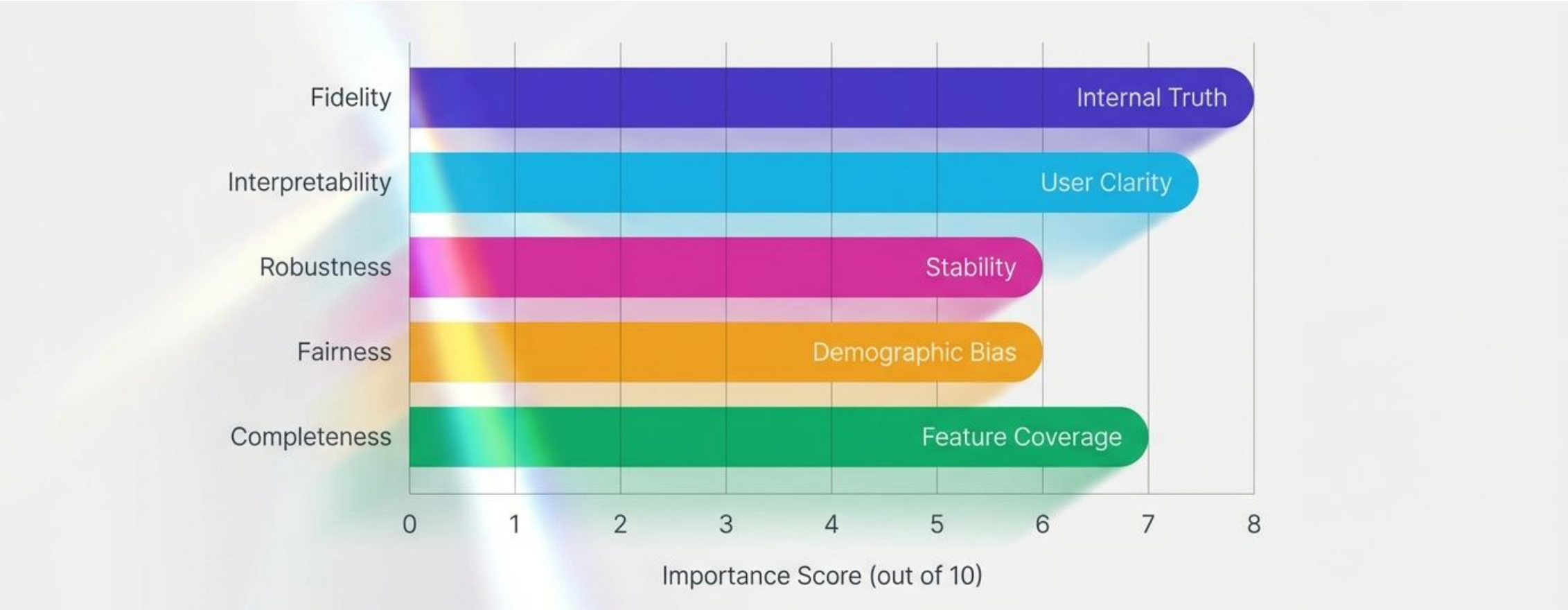
"Moving from 'Does it work?' to 'Does it work for the right reasons?'"

Why current metrics fail to agree on “faithfulness”.



Different metrics rank the same explanation differently. Without a unified framework, framework, evaluation is subjective.

Evaluation Dimension



Dimension 1: Fidelity

Ensuring the explanation aligns closely with the model's actual decision-making process.

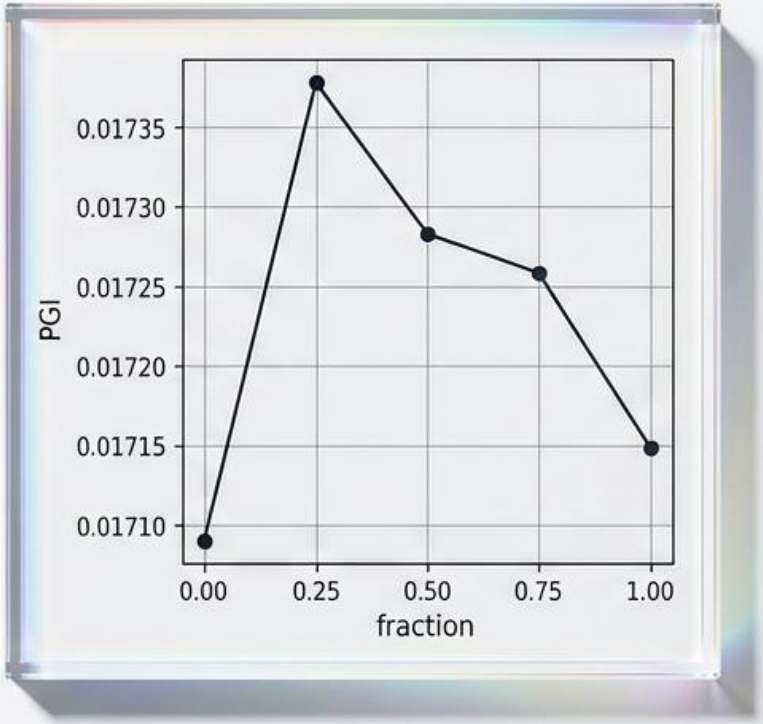
Litmus Test:

If the explanation says a feature is important, changing that feature MUST change the prediction.

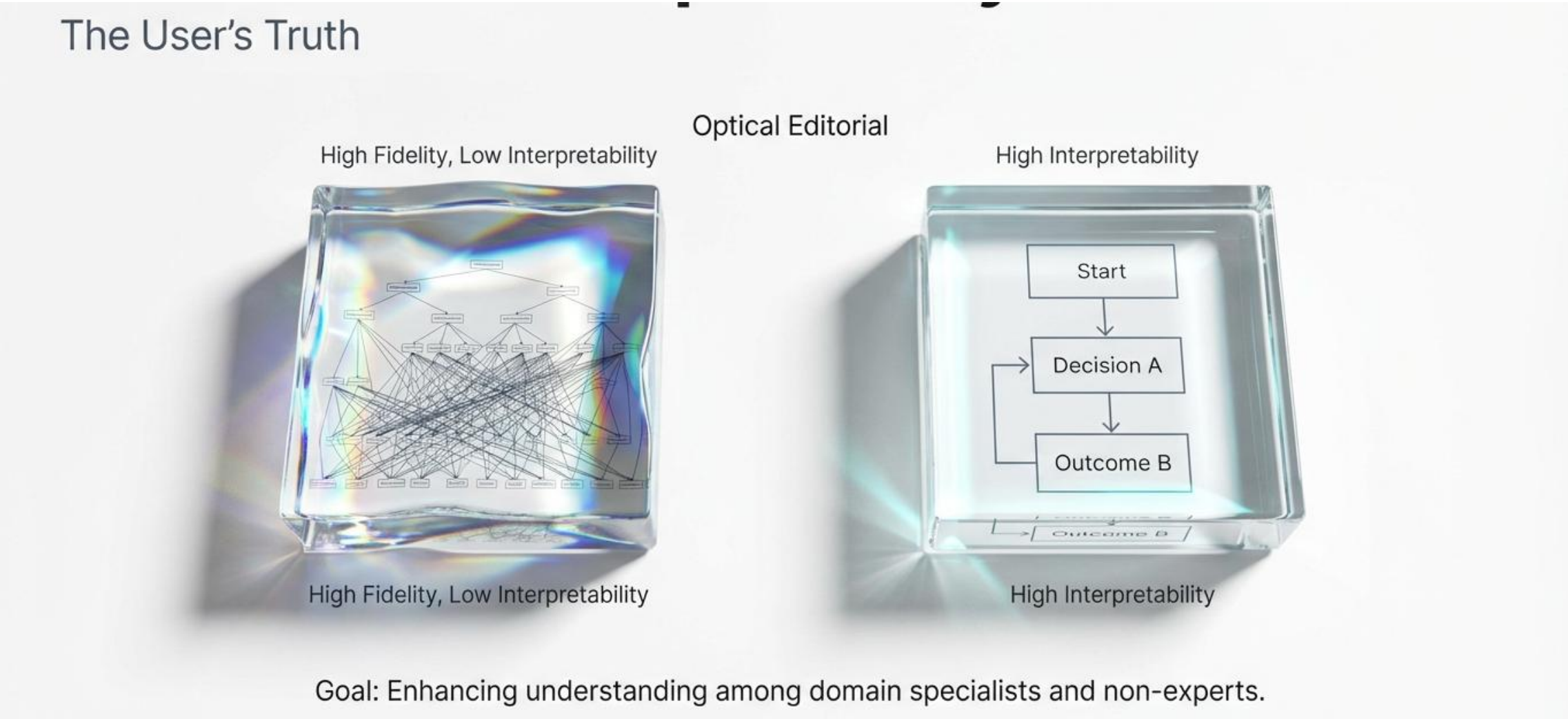
Measuring Fidelity: PGI metric

$$PGI = \frac{1}{m} \sum |f(x) - f(\tilde{x})|$$

Identify top k features → Perturb (scramble) features → Measure prediction drop

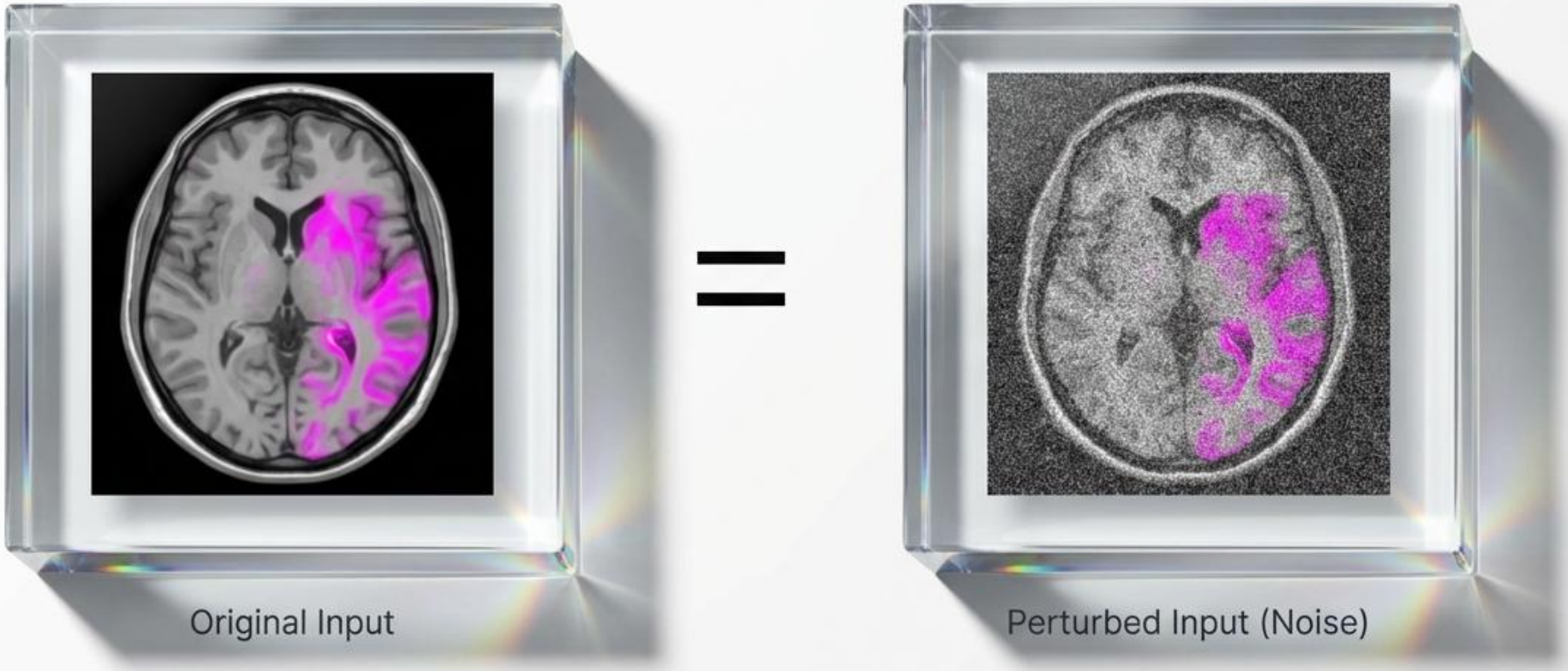


Dimension 2: Interpretability



Dimension 3: Robustness

Stability Against Noise



A robust explanation must remain consistent even when the input is perturbed by noise or adversarial attacks.

The spurious corellation trap

Understanding Hidden Biases

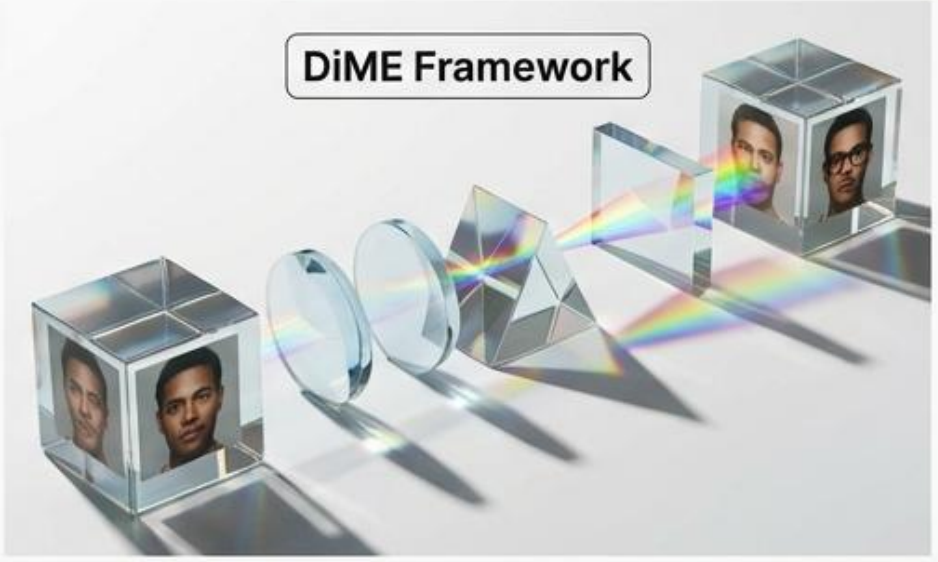
The Model's Shortcut

Problem: Standard metrics like MNAC penalize explanations that identify true causal links if those links involve multiple attributes (e.g., Lipstick + Gender = Age prediction).

We need metrics that detect 'cheat codes'.

Solving for spuriousness: Corellation difference (CD)

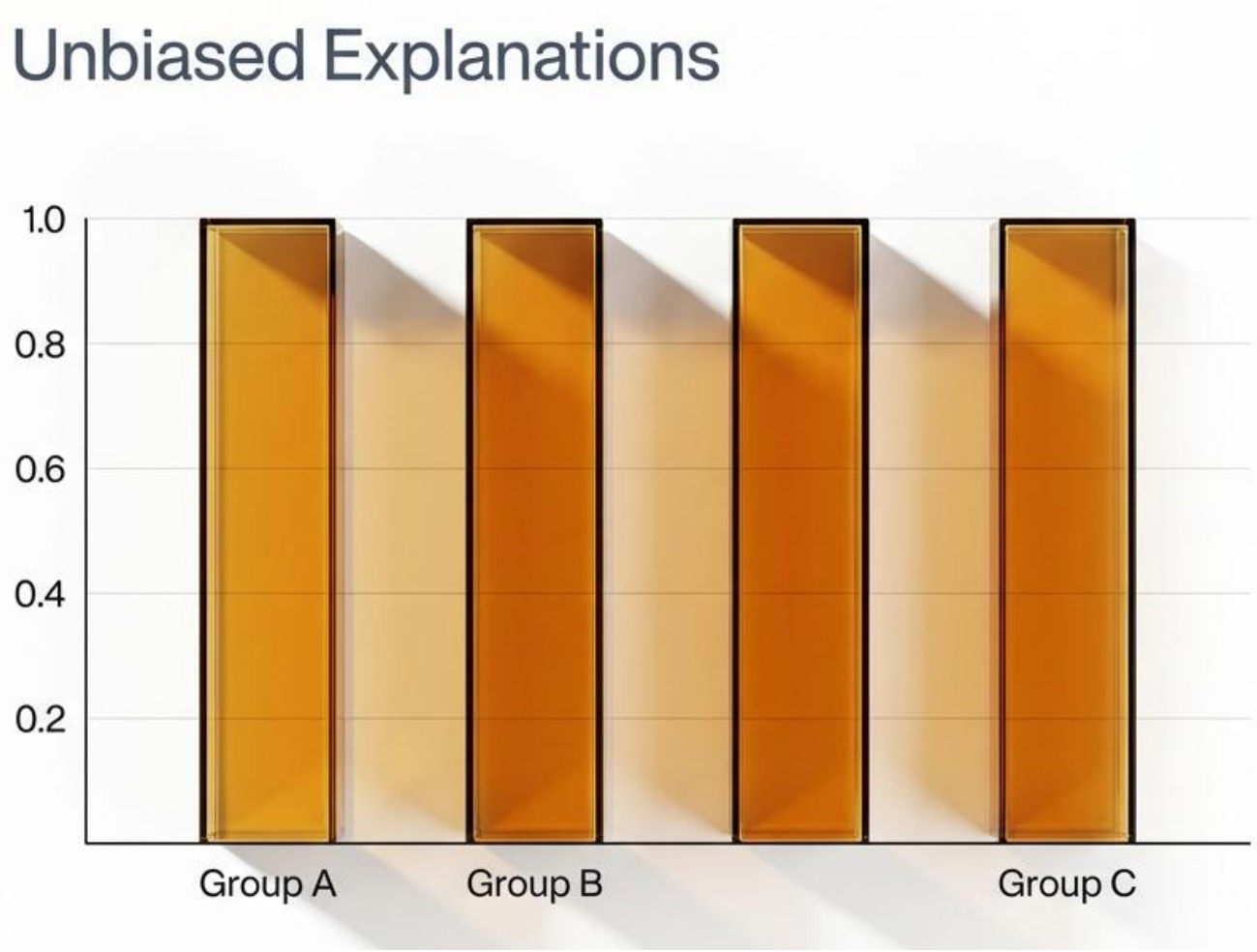
$$CD_q = \sum |c_{q,a} - c^M_{q,a}|$$



CD measures how well the explanation captures the *true* data correlations versus the model's learned shortcuts. Low Score = High Trust.

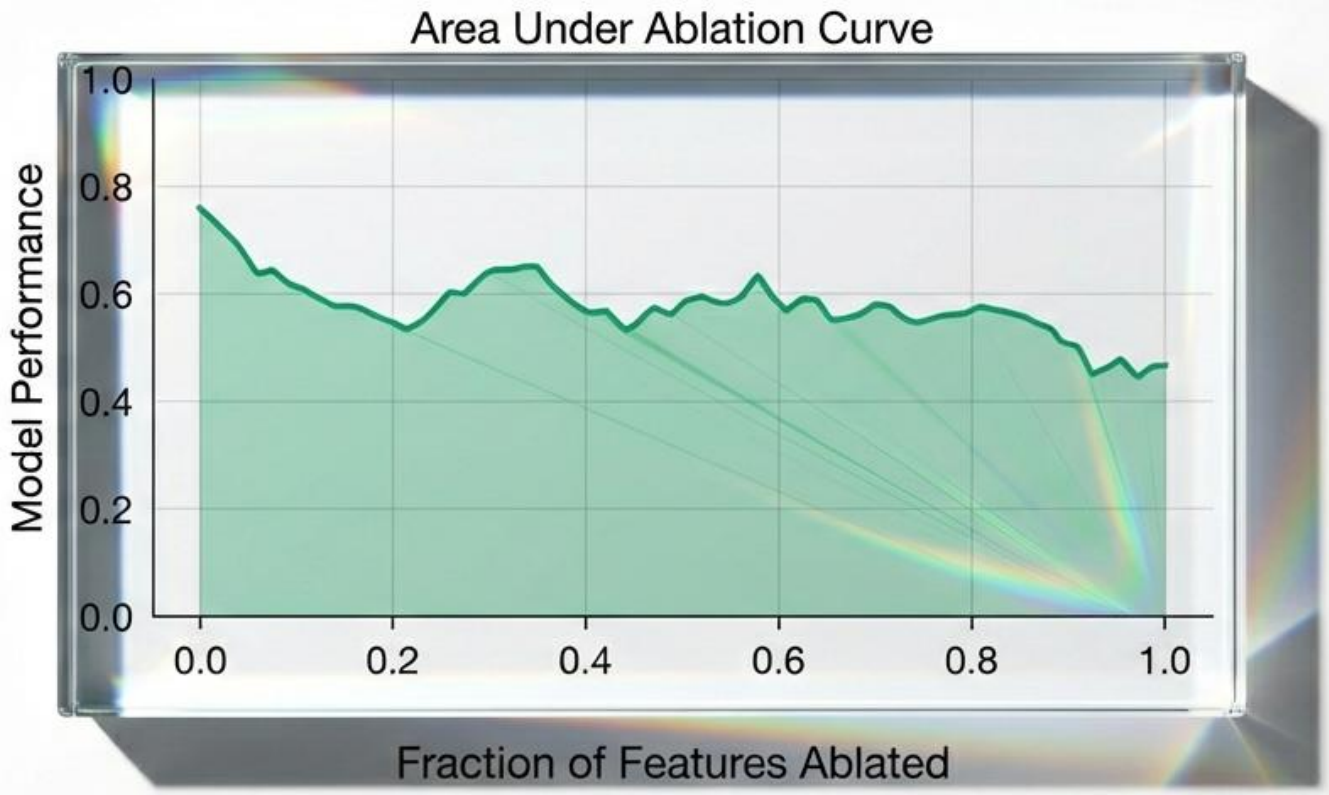
NotebookLM

Dimension 4: Fairness



Assessing the neutrality of explanations among demographic groups. Critical for Finance and HR.

Dimension 5: Completeness and Ablation



Method: Systematically remove “unimportant” features. If accuracy drops, the explanation missed something.

Dynamic Weighting...

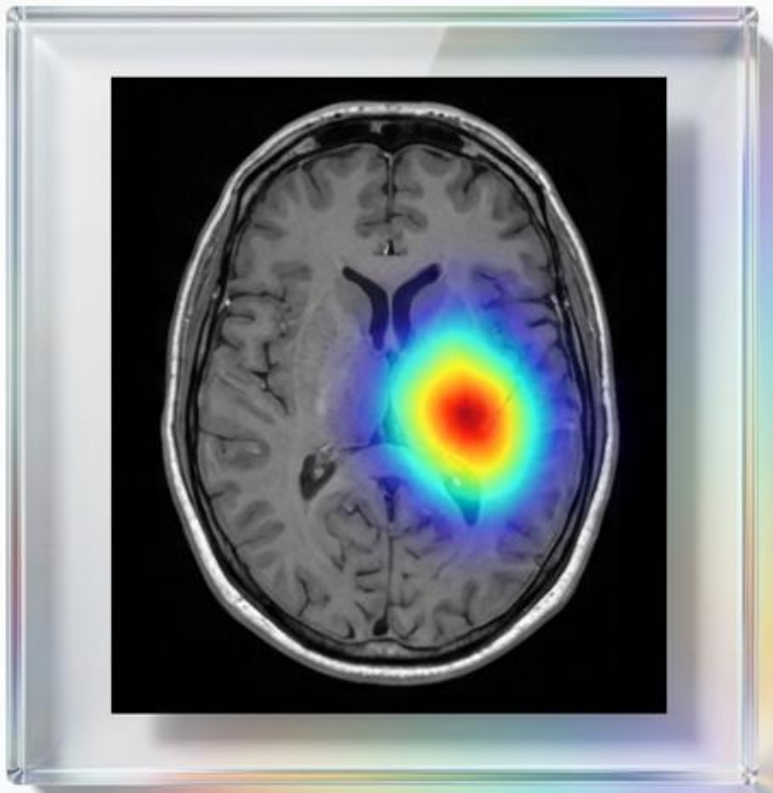
$$S = \sum w_i \cdot s_i$$

One size does not fit all. We adjust the weights (w_i)
We adjust the weights (w_i) based on industry priorities.

Domain Focus: Healthcare

Priority: Interpretability (30%) + Completeness (25%)

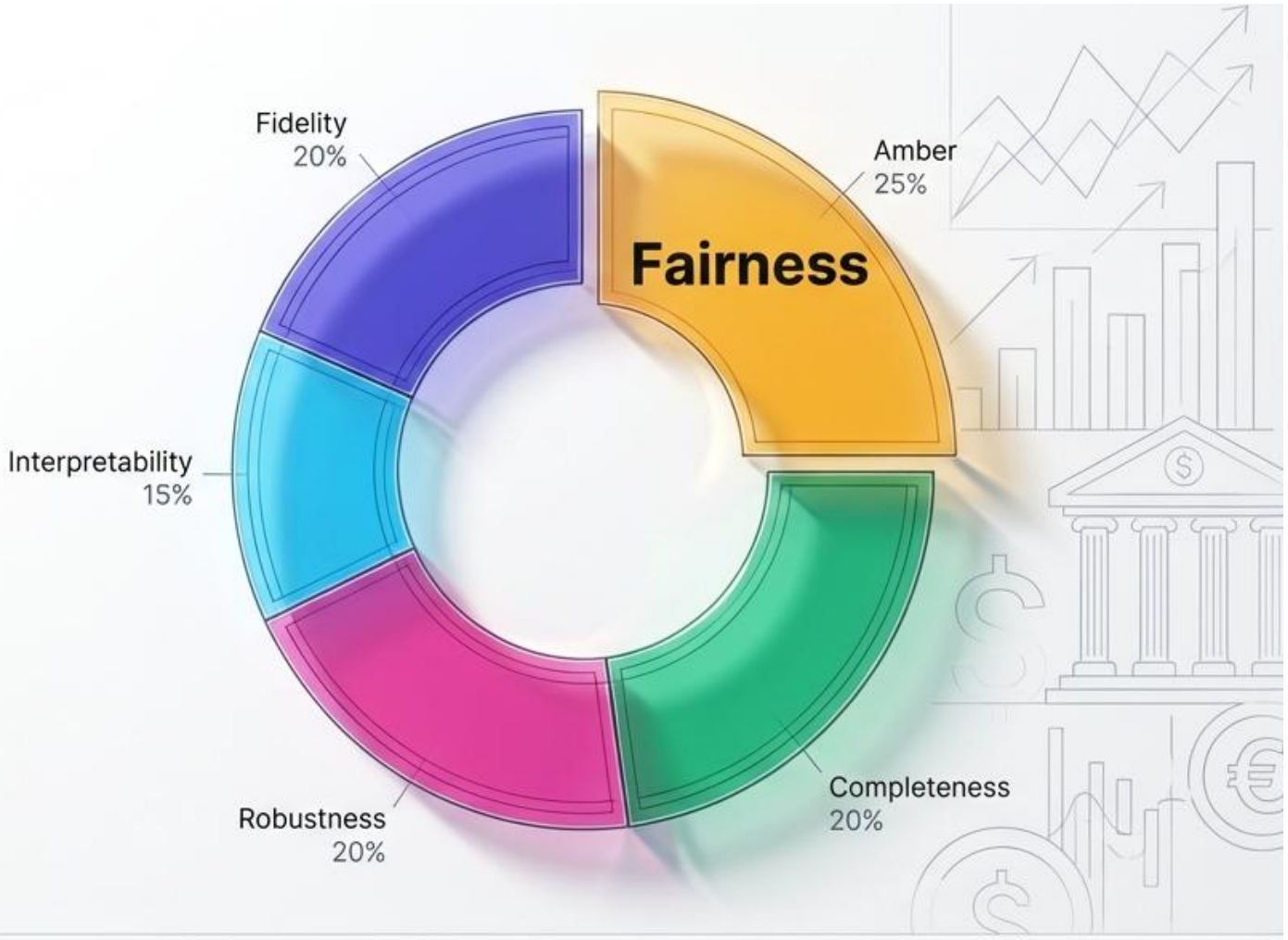
Why: Clinician Trust & Liability. A doctor must know exactly why a diagnosis was made.



Finance?

Priority: Fairness (25%)

Why: Regulatory Compliance. Proving lack of bias in loan approvals (e.g., Fair Credit Reporting Act).



Security?

Priority: Robustness (15%) + Fidelity (25%)

Why: Adversarial Defense. Detecting forbidden items despite noise or concealment attempts.

Human-Center Evaluation

Shift from Computer-Centered to Human-Centered.

Goal: Does the AI's reasoning match the Expert's reasoning?

Method: User feedback loops and cognitive process tracing.

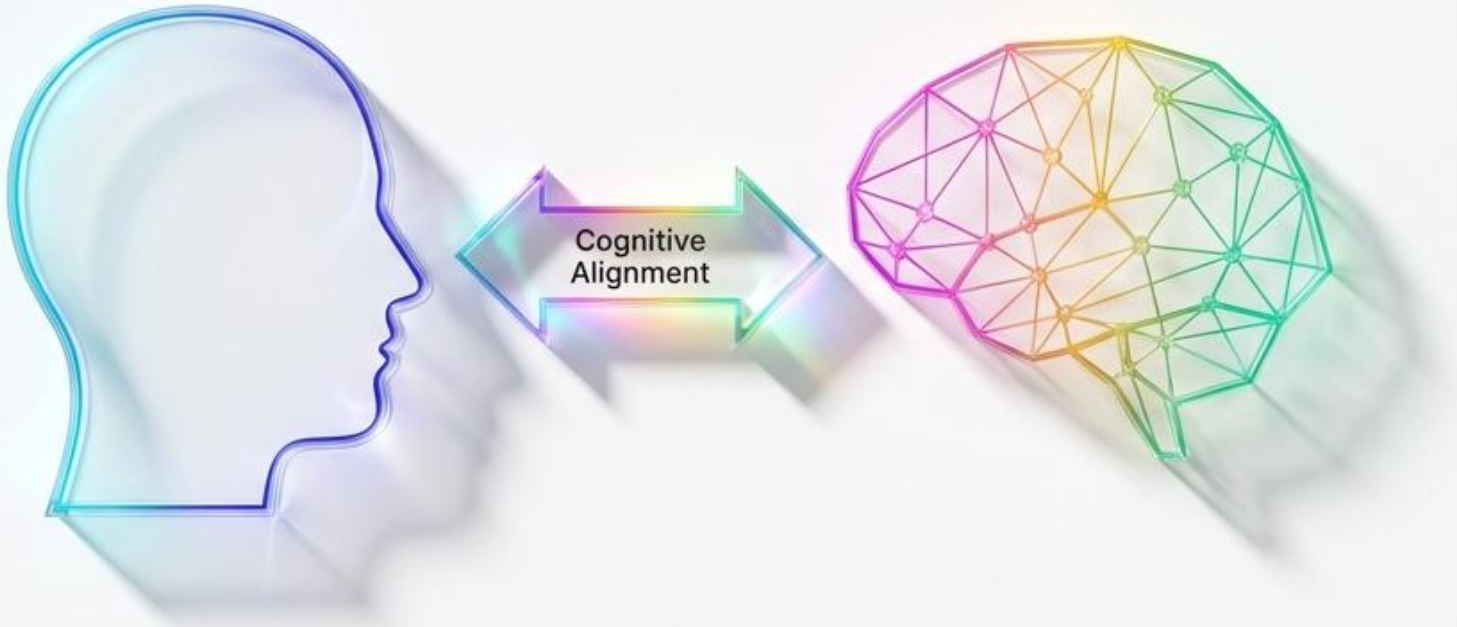


Illustration of Understanding

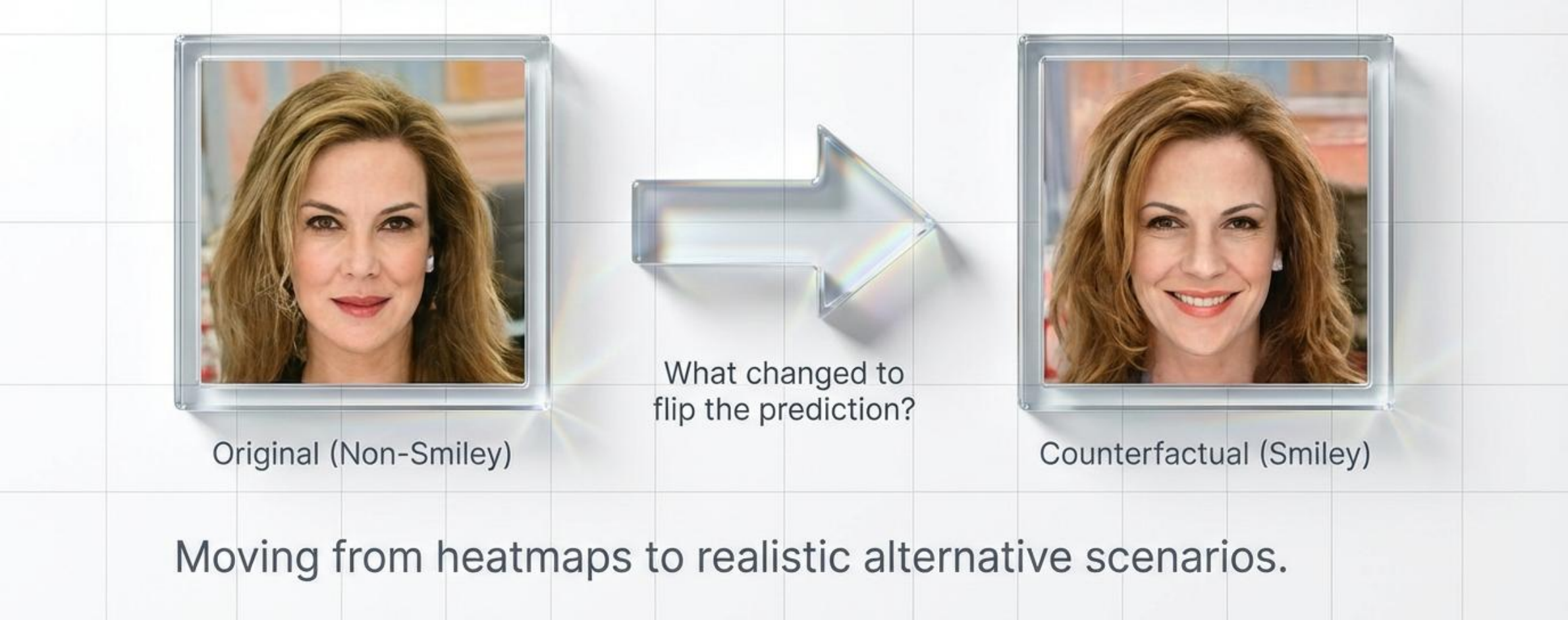
Critique: Saliency maps often act as simple edge detectors. Humans project their own reasoning onto ambiguous blobs.

Benchmarking the frameworks

A comparative analysis across key dimensions of trust.

	Fidelity	Interpretability	Robustness	Fairness	Scalability
LIME	◐	●	○	○	◐
SHAP	●	◐	○	○	◐
Grad-CAM	◐	◐	○	○	◐
Proposed Framework	◐	◐	●	●	●

Generetive Explaination



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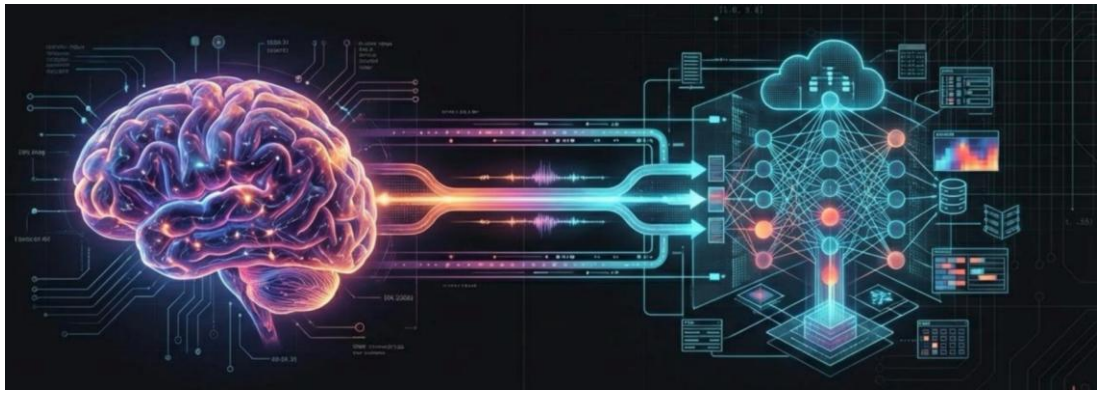
Thanks